

Multi-Law-Firm Architecture: Inbound Routing, AI Isolation & Multi-Clio Integration

How a Single Voice Agent Platform Simultaneously Serves Multiple Independent Law Firms with Strict Data Isolation

Target Platform: Multi-Tenant Voice Agent Core <—> Independent Clio Accounts

Architecture Mode: Multi-Tenant SaaS / Agency Provider

1. Executive Summary: The Multi-Firm Challenge

When running a Voice Agent platform for **multiple different law firms** (e.g., *Apex Legal Group*, *Bottini Law*, and *Gotham Partners*), the system must solve three mission-critical problems:

- 1. Inbound Identification:** When a phone call arrives, how does the Voice Agent know which law firm is being called?
- 2. AI Brand & Persona Isolation:** The agent must greet the caller with the correct firm name, attorney roster, and practice areas.
- 3. Data Isolation & Multi-Clio Dispatch:** The caller's contact, transcript, and legal matter must route *strictly* to that specific firm's Clio account, with zero risk of Firm A seeing Firm B's clients.

Core Architectural Principle: The **Dialed Phone Number (DID)** is the universal routing key. Every law firm owns a distinct inbound phone number. When a call arrives, the Voice Agent matches the dialed number against a multi-tenant registry, loads that firm's specific AI prompt, and selects that firm's dedicated Clio OAuth token for post-call sync.

2. How the Voice Agent Identifies the Law Firm (DID Matching)

In enterprise telephony, every phone number is known as a **Direct Inward Dialing (DID)** number. Whether a firm uses Twilio, Telnyx, or LiveKit SIP trunks, the carrier's incoming SIP INVITE always contains two essential headers:

- From :** The caller's personal number (e.g., `+1-555-019-3388`).
- To :** The dialed phone number that belongs to the law firm (e.g., `+1-555-222-0001`).

The Tenant Phone Registry Mapping Table

Firm Identifier	Assigned Inbound Number (DID)	Firm Name & Brand	Target Clio Manage Account
<code>firm_apex</code>	<code>+1 (555) 234-5678</code>	Apex Legal Group	Clio Account ID: <code>US-99120</code> (Token: <code>41369-XSRy...</code>)
<code>firm_bottini</code>	<code>+1 (555) 888-4321</code>	Bottini Law	Clio Account ID: <code>US-44812</code> (Token: <code>77102-MKQz...</code>)
<code>firm_gotham</code>	<code>+1 (555) 999-7700</code>	Gotham Partners	Clio Account ID: <code>US-11509</code> (Token: <code>99341-PL0a...</code>)

3. End-to-End Multi-Firm Workflow Diagram

The diagram below demonstrates how two callers dialing different law firms at the exact same second are handled concurrently without data cross-contamination:



4. Multi-Tenant Token Vault & Database Design

In a multi-firm architecture, your Voice Agent database maintains an isolated `organizations` and `integrations` table:

```
{
  "tenants": {
    "org_apex_01": {
      "firm_name": "Apex Legal Group",
      "inbound_phone": "+15552345678",
      "ai_greeting": "Thank you for calling Apex Legal Group.",
      "practice_areas": ["Personal Injury", "Car Accidents"],
      "clio_integration": {
        "connected": true,
        "clio_account_id": "99120",
        "access_token": "41369-XSRyHVforKl0AK2o2wTys4P3Na61gjkbJM",
        "refresh_token": "ncyIYwVfVA9pLH0erXI1yaZ8U5a0BUmSqysZ3Irm",
        "token_expires_at": 1792748400
      }
    },
    "org_bottini_02": {
      "firm_name": "Bottini Law",
      "inbound_phone": "+15558884321",
      "ai_greeting": "Thank you for calling Bottini Law.",
      "practice_areas": ["Securities Litigation", "Class Action"],
      "clio_integration": {
        "connected": true,
        "clio_account_id": "44812",
        "access_token": "88192-PLKxYJqoRmK1BL3o3wUzs5Q40b72hkclKN",
        "refresh_token": "mdxHZwUeUB8oKG9dqWH0xzY7T4z9ATmRPxrY2Hql",
        "token_expires_at": 1792751200
      }
    }
  }
}
```

5. How Each Law Firm Connects Their Clio Account (The OAuth Flow)

You do not need to manually collect passwords from your law firms. Each firm connects securely via Clio OAuth 2.0:

Step 1: Send the Firm Their Unique Onboarding Link

When you onboard *Bottini Law*, your platform generates a custom authorization link containing a unique `state` parameter:

```
https://app.clio.com/oauth/authorize?response_type=code&client_id=1zYn1AJ48HtBvdTZz300LljEqPKYhHQZKAWK11tz&redirect_uri=ht
```

Step 2: Managing Partner Clicks "Allow Access"

The managing partner at Bottini Law logs into their own Clio account and clicks **Allow Access**.

Step 3: Callback Stores Tokens to That Specific Firm

Clio redirects back to your dashboard with both the authorization code and the state: `/oauth/callback?code=AUTH_CODE&state=org_bottini_02`. Your server exchanges the code and saves the resulting token under `org_bottini_02`.

6. Adding Firm #2, Firm #3, etc. (Step-by-Step Operator Checklist)

Operator Checklist for Onboarding a New Law Firm:

Step 1: Create the Organization in Voice Agent:

Go to your Voice Agent dashboard at `/organizations` and click *Create Organization* (e.g., "Bottini Law").

Step 2: Assign a Dedicated Phone Number:

Go to *Telephony > Buy Phone Number* or assign an existing DID to that organization.

Step 3: Configure Practice-Specific AI Prompt:

In Agent Builder, set the firm's custom greeting, business hours, and intake questions.

Step 4: Connect the Firm's Clio Account:

Generate their unique OAuth connect link. Once authorized, all calls to their dedicated phone number automatically route into their private Clio account!

7. Security, Legal Ethics & Malpractice Prevention

- **Zero Data Bleed (Strict Multi-Tenancy):** Every database query and Clio API dispatch strictly filters by `workspace_id`. There is no shared memory between active calls.
- **Independent Conflict Checking:** When an intake call arrives for Firm A, the conflict check only searches Firm A's Clio database, ensuring strict compliance with attorney-client confidentiality and ABA Model Rule 1.6.
- **Independent Client Portals:** When Firm A invites Mary, she sees Firm A's logo and legal team in her portal. Firm B's clients see Firm B's branding.